BEREKET TILAHUN

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Experience

DZ Software Engineering PLC **Jul 2023 – Present**

Senior Software Engineer *Addis Ababa, Ethiopia*

Promoted from Junior Software Engineer after 16 months. Completed my BSc in Computer Science at HiLCoE alongside this role.

- Built and operated **15 containerized backend services** on **FastAPI** and **Django REST**, packaging async Python workloads with Docker and shipping through CI/CD; layered **Redis caching** (84% hit rate) to drop p95 latency from **340ms to 195ms** at **420 req/min** peak.
- **Architected an LLM evaluation pipeline** and CI gate that benchmarks correctness, reliability, and runtime across **Python, JavaScript, Java, C++, and Rust** submissions; rolled it out as an automated quality gate on **11 production services**, cutting manual review by **40%** with **zero correctness regressions** since launch.
- Designed an **automated benchmark curation pipeline** over **PostgreSQL** telemetry and **Redis** event streams that scores Python, JS, Java, C++, and Rust traces on latency, correctness, and reliability—using Rust as the sub-millisecond baseline—reducing **MTTD by 30%** and producing **1,200+ labeled records** for fine-tuning.
- Hardened **47 API endpoints** with schema-enforced validation across Python and Java, parameterized queries, rate limiting, and **Bandit / pip-audit** pre-merge gates—closing **9 High/Critical CVEs** with no recurrence in the cycles that followed.
- **Led a Zero Trust migration**: replaced **11 long-lived secrets** with short-lived, scope-limited **OIDC** tokens, made **Trivy** image scans a hard deploy gate, and stood up an append-only audit pipeline—dropping incident reconstruction to **under 20 minutes** with **zero auth-related production incidents** to date.
- Deployed evaluation services on **AWS** (EC2, S3, IAM role-scoped access, VPC-isolated mesh) behind load-balanced ingress, achieving environment parity across **11 staging/production pairs** and reducing rollback time by **35%**.

DZ Software Engineering PLC **Mar 2022 – Jun 2023**

Junior Software Engineer *Addis Ababa, Ethiopia*

- Triaged **10–12 Tier 1 incidents daily** across Python, Java, and C++ services; authored runbooks and automated recurring health checks, cutting recurring failures by **15%** across enterprise environments.
- Diagnosed and resolved OS, network, and distributed-service failures end-to-end, establishing observability baselines that were later adopted across **3 production environments**.

Remote (Independent) **Mar 2021 – Jun 2023**

Freelance Backend Engineer (part-time) *Addis Ababa, Ethiopia*

Client work I took on before joining DZ, and continued part-time alongside it until mid-2023.

- Shipped **15+ production backend systems** spanning AI evaluation, collaboration, and security on **FastAPI**, **Django REST**, and **Spring Boot**, with **React / Node.js** frontends—serving **100+ concurrent users** at sub-200ms API targets.
- Engineered a **multilingual code search backend** with **BM25** and **TF-IDF** retrieval over Python, Java, C++, and Rust corpora; enforced authorization-before-query checks to neutralize timing-oracle attacks and applied **AES-256** tenant-scoped encryption at rest.
- Built a **distributed real-time evaluation platform** using **PostgreSQL** row-level optimistic locking, async Python orchestration, and a **WebSocket** notification layer—hitting a **79% cache hit rate** and pulling p95 latency from **210ms down to 48ms** under 85 concurrent connections.
- Designed an **agent-assisted evaluation pipeline** using async Python workers and **LLM-generated test scaffolds** that spans the full lifecycle—API contract checks, implementation correctness, and production monitoring—across **12 services**, lifting throughput by **18%**.
- Closed the loop on full-stack quality by folding correctness scoring, latency profiling, and reliability checks into a unified **CI/CD-gated pipeline**, eliminating manual handoffs between frontend, API, and backend stages.

Skills

Languages: **Python**, **JavaScript**, **Java**, **C++**, **Rust**, C#, Dart, SQL
AI & Evaluation Systems: **LLM Evaluation Pipelines**, Multi-Language Code Benchmarking, **Prompt Engineering**, Correctness & Reliability Verification, Agent-Based Validation, Performance Benchmarking, Fine-Tuning Data Curation
Backend & Architecture: **FastAPI**, Django REST, Spring Boot, Node.js, REST API Design, Async Processing, **Microservices**, Event-Driven & Distributed Systems
Cloud & DevOps: **AWS** (EC2, S3, IAM, VPC, ALB), **Docker**, **CI/CD**, Linux, Git, Trivy, Bandit, Observability
Data Systems: **PostgreSQL**, **Redis**, SQL Server, BM25 / TF-IDF Retrieval
Security Engineering: **Zero Trust**, **OIDC**, Keycloak, JWT, RBAC / ABAC, OWASP Top 10, HMAC, Token Lifecycle Design

Education

Bachelor of Science in Computer Science **Jul 2026**

HiLCoE School of Computer Science & Technology *Addis Ababa, Ethiopia*

Completed while working as a software engineer at DZ Software Engineering (2022–2026).